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DATA 612

Research Discussion Assignment 1

Question 1

Now that we have covered basic techniques for recommender systems, choose one commercial recommender and describe how you think it works (content-based, collaborative filtering, etc). Does the technique deliver a good experience or are the recommendations off-target?

You may also choose one of the three non-personalized recommenders (below) we went over in class and describe the technique and which of the three you prefer to use.

1. Metacritic: How We Create the Metascore Magic
2. **Rotten Tomatoes: About Rotten Tomatoes**
3. IMDB: FAQ for IMDb Ratings

Netflix Recommendation System

- I believe Netflix uses a hybrid recommendation approach that combines collaborative filtering and content-based filtering.
 - Collaborative filtering comes in as it identifies users with similar viewing habits and recommends content that those users enjoyed.
 - Content-based filtering recommends movies and shows that share characteristics with content a user has previously watched, such as genre, actors, directors, or themes.
- Overall, I think Netflix provides a very good user experience. Most of the recommendations are relevant to my interests and often help me discover content I would not have searched for on my own. However, the recommendations can sometimes become repetitive because the system tends to recommend content similar to what I have already watched.

Rotten Tomatoes Recommendation System

- Among the three non-personalized recommenders discussed, I prefer to use Rotten Tomatoes.
- Unlike Netflix, Rotten Tomatoes does not personalize recommendations for individual users.
 - Instead, it aggregates reviews from professional critics and audiences.
 - The Tomatometer represents the percentage of critic reviews that are positive, while the Popcornmeter reflects audience sentiment.
 - * A movie is considered “Fresh” when at least 60% of critic reviews are positive.
 - * The Popcornmeter similarly summarizes audience ratings and reviews.
- I like Rotten Tomatoes because it provides a quick way to gauge whether a movie is generally well-received. While it does not account for my personal preferences, it is useful for comparing movies and deciding whether something is worth watching.

Netflix focuses on recommending content specifically for me based on my viewing history, while Rotten Tomatoes provides a general recommendation based on the opinions of critics and audiences. Personally, I find Netflix more useful for discovering new content, but I often check Rotten Tomatoes before deciding whether to invest my time in a movie or TV show.

Question 2

Read the article below and consider how to handle attacks on recommender systems. Can you think of a similar example where a collective effort to alter the workings of content recommendations have been successful? How would you design a system to prevent this kind of abuse?

Travis M. Andrews, The Washington Post (2017): Wisdom of the crowd? IMDb users gang up on Christian Bale’s new movie before it even opens.

Attacks on Recommender Systems

The article highlights how recommender and rating systems can be manipulated when large groups of users coordinate their actions. In the example discussed, people intentionally submitted negative ratings for a movie before it was even released, influencing how the movie appeared to potential viewers. This demonstrates how crowd-sourced recommendation systems can be vulnerable to abuse when ratings are based on factors other than actual user experience.

A similar example is when someone becomes involved in a social media controversy and large groups of people leave negative reviews of their employer, business, or workplace, even though they have never interacted with the company. These coordinated efforts can significantly lower

ratings and influence recommendation systems that rely on user reviews. In these situations, the goal is often to damage a person's or organization's reputation rather than provide honest feedback about the quality of the service or product.

To prevent this type of abuse, I would design the system to prioritize reviews from verified users. For example, only customers who have actually purchased a product, used a service, or interacted with a business would have their reviews fully counted in recommendation calculations. I would also implement anomaly detection to identify unusual patterns, such as a sudden spike in negative reviews from new accounts or users with no previous history. Reviews flagged as suspicious could be temporarily excluded until they are verified.

I think combining verified reviews with automated monitoring would help ensure that recommendations reflect genuine user experiences rather than coordinated attempts to manipulate ratings.